

Task 1: Order Flow Imbalances (OFI)

1. What's the motivation behind measuring OFI at multiple depth levels of the order book?

The motivation behind measuring Order Flow Imbalance (OFI) across multiple depth levels is to capture richer and more informative signals about market pressure and liquidity than what best-level OFI alone can provide. Here's the rationale:

- Best-level OFI is limited to the top of the order book, which may not fully reflect trading intentions, especially in volatile or less liquid markets.
- Deeper levels of the order book often hold substantial hidden liquidity, and changes at these levels can anticipate large trades or strategic order placements.
- Price impact is not limited to top-level interactions. Especially during large market moves, orders several levels deep influence the price dynamics.

Empirical Justification from my Results(Python Code):

From our Lasso regression analysis:

- Best-Level OFI showed an R^2 of 0.1729 on the test set.
- In contrast, Multi-Level OFI achieved a significantly higher R^2 of 0.3675, indicating better explanatory power for log returns.
- This means multi-level OFI captured more variance in price movements, validating that deeper order book information is materially useful in modeling price impact.

These results clearly support the argument that including OFIs from multiple levels improves predictive performance, and thus, justifies the need for measuring OFI beyond the best level.

2. Why do the authors use Lasso regression rather than OLS for estimating cross-impact?

The authors opt for Lasso regression (Least Absolute Shrinkage and Selection Operator) over Ordinary Least Squares (OLS) for several practical and statistical reasons when estimating cross-impact models, particularly in high-dimensional, intraday market settings:

1. High-Dimensionality and Ill-Posed Nature of OLS:

OLS becomes ill-posed when the number of predictors exceeds the number of observations, leading to overfitting and unstable coefficient estimates. In cross-impact modeling, especially with approximately 100 variables (assets or depth levels), and limited intraday observations (e.g., a 30-minute window with 1-minute returns yields only 30 observations), OLS is not feasible.

2. Multicollinearity Among Predictors:

OLS requires predictors to be linearly independent. However, multi-level OFIs and cross-asset OFIs exhibit significant multicollinearity, as shown in Figure 3 of the referenced paper. For example, 10% of the cross-asset OFI correlations exceed 0.30, violating OLS assumptions.

3. Sparsity in Cross-Impact Relationships:

It is empirically observed (e.g., Capponi & Cont, 2020) that only a subset of variables meaningfully influence the price of a stock. Lasso naturally enforces sparsity by shrinking irrelevant coefficients to zero, which helps in both feature selection and interpretability.

4. Empirical Validation from my Results (Python Code):

In our implementation:

- Lasso regression on multi-level and integrated OFIs yielded interpretable and stable results.
- Only a few coefficients were non-zero, confirming the sparse nature of impactful predictors and validating the authors' modeling assumption in practice.

3. Why is OFI considered a better predictor of short-term returns than trade volume?

Order Flow Imbalance (OFI) captures the net pressure between buying and selling interests at various levels of the limit order book. This provides a direct representation of supply and demand dynamics, which is more informative than trade volume alone for the following reasons:

1. Trade Volume is Directionless:

Trade volume only tells us how much trading occurred, not who initiated the trade (buyer or seller). In contrast, OFI distinguishes between aggressive buying and selling, making it a directional signal.

2. OFI Reacts to Order Book Updates:

OFI is updated tick-by-tick using changes in bid/ask prices and sizes, offering finer granularity than aggregated trade volume. This allows it to capture immediate liquidity imbalances that precede short-term price changes.

3. Reflects Latent Market Pressure:

Large market orders or cancellations at the top levels cause shifts in OFI before any actual trades happen. Thus, OFI anticipates price movements even in the absence of trades, while volume responds after the fact.

4. Empirical Evidence from Lasso Analysis (Python Code):

- Lasso regressions using OFI (Best-Level, Multi-Level, Integrated) showed significant R^2 values (e.g., 0.3675 for Multi-Level OFI).
- These results suggest that OFI features explain a notable portion of return variance, whereas volume alone would not offer the same predictive power.

Conclusion: OFI offers a more timely, directional, and granular view of market dynamics than trade volume, making it a superior predictor of short-term price movements.